

if

WORLDS OF SCIENCE FICTION

MAY 1955

35 CENTS



SNOWBALL

By Poul Anderson

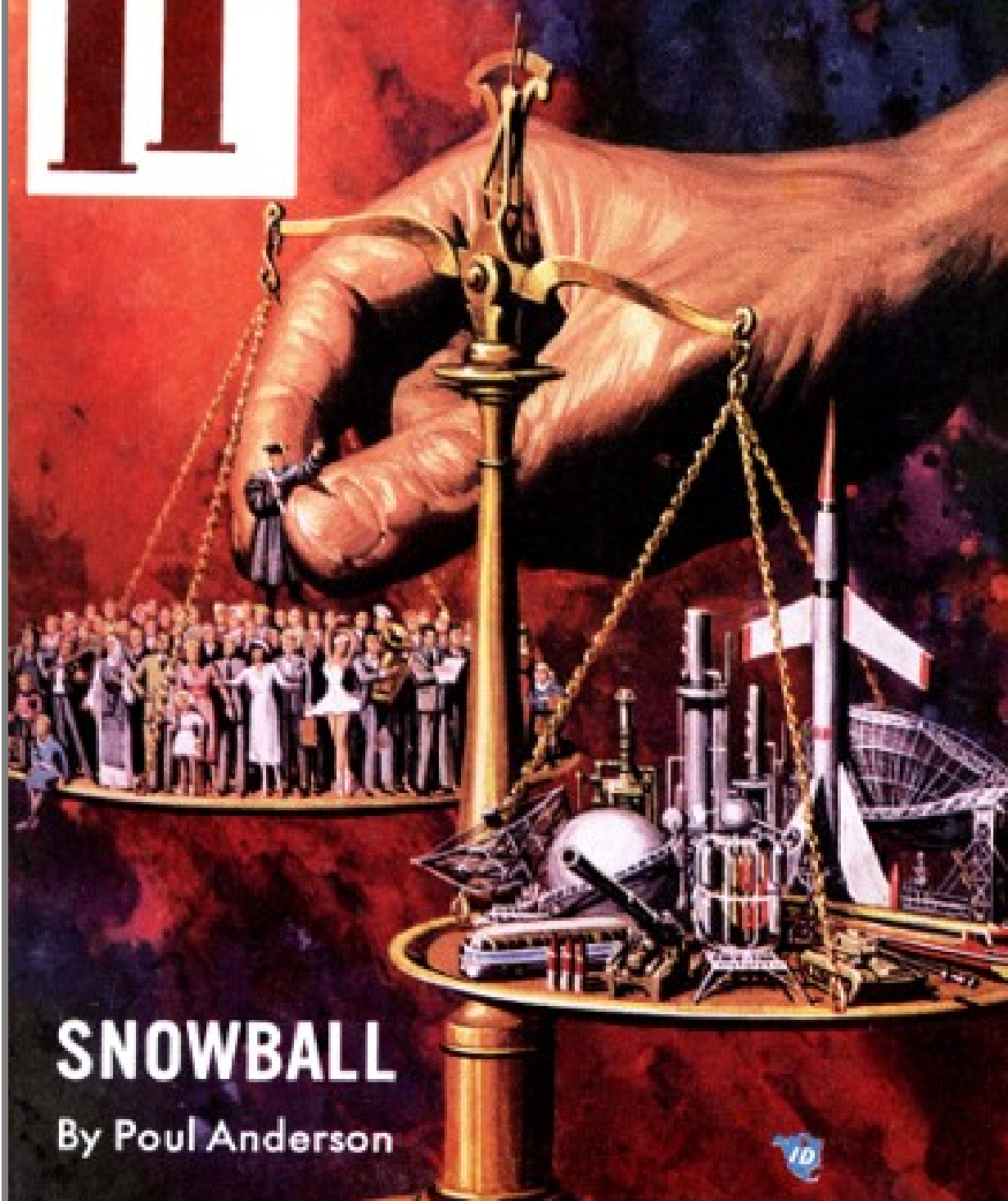


if

WORLDS OF SCIENCE FICTION

MAY 1955

35 CENTS



SNOWBALL

By Poul Anderson



The Project Gutenberg eBook of The Laboratorians

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: The Laboratorians

Author: Edward Peattie

Illustrator: Ed Emshwiller

Release date: April 13, 2019 [eBook #59267]

Language: English

Other information and formats: www.gutenberg.org/ebooks/59267

Credits: Produced by Greg Weeks, Mary Meehan and the Online Distributed Proofreading Team at <http://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK THE
LABORATORIANS ***

The Laboratorians

BY EDWARD PEATTIE

Playing "Napoleon" can get to be a habit, especially when a man is devoted to pure science. Which was Dr. Whitemarsh's devotion—until Dr. Sally Chester came along!

**[Transcriber's Note: This etext was produced from
Worlds of If Science Fiction, May 1955.
Extensive research did not uncover any evidence that
the U.S. copyright on this publication was renewed.]**

"Yeah, we drop in just three c.c. from this here tube," said Rocco as he expertly twirled the erlenmeyer flask and watched the color shoot past the methyl orange end-point. Whitemarsh was annoyed and said so.

"That's the sixth straight you've missed, and the acid comes out of the burette, not the tube; and you don't call the graduations c.c., you call them milliliters."

"Yeah? Well, here we call it a tube!"

"And why don't you go down to the end-point drop by drop?"

"Because the book don't say so! That's why! You technos make me sick. Here we do all the blasted work, and you try to tell us how to do what we've been doing for ten years!"

Rocco was beginning to work himself into one of his famous rages. His bull neck was beginning to redden; his eyes started to flash. His entire squat body started to quiver.

Whitemarsh wasn't impressed. Over at the atomic plant, Phobus's Quercus Mountain, he had bossed a pretty quarrelsome crew of isotope wranglers. He had never dodged a fight in his life. But this was in a chemical laboratory and it surprised him to hear the assistants talk back.

The only assistants he had ever known were clear-eyed youths taking a year away from their studies to recoup their tuition money and who tried to copy everything the chemists did. But Whitemarsh was new to the Interspatial Research Center on the Moon, and he still could not figure why the assistants acted as they did. So he waited.

Rocco banged the flask down on the stone bench, glared at Whitemarsh for an instant, and then rushed out of the Laboratory, muttering a few obscenities.

"Queer place this," mused Whitemarsh, filling up another flask and finishing the titration himself. "Here the helpers tell the chemists what to do and get mad if we ask them what they're doing."

He started to look over Rocco's notes and ruefully decided all the work would have to be done over again. He was interrupted when a girl opened

the door. In the week he had been stationed at IRC, he had been introduced to so many scientists that he had forgotten most of the names, but he remembered all the girls. His former Atomic Plant at Quercus Mountain had had all too few for him not to appreciate them now. Miss Sally Chester was a statuesque chemist with long blonde hair and a luscious figure which she hid under a white lab robe. He managed to stammer some sort of greeting.

"Why Dr. Whitemarsh!" She seemed somewhat puzzled. "You're not actually working with your hands?"

"I sure am, unless we're both space struck. Why not?"

"Well, I suppose it's all right other places, here we let the Laboratorians do all the manual work. It's sort of their privilege."

"Yes, but their technique's lousy. I sat here this afternoon and watched that blow-hard Rocco muff six straight end-points in a row and when I asked him how come, he blew his top!"

She laughed at that. She sat down on the lab desk and said, "You're absolutely right. Antonio Rocco's color blind and always misses his Methyl Orange end-points. And he's been doing them for ten years. But it hurts his feelings to be criticized, you should have been more diplomatic. He's probably gone to complain to his boss!"

"His boss? Aren't we his bosses? On this sheet he's listed as my assistant."

"Actually yes. But traditionally the shop foreman is the leader of the Laboratorians. He certifies them to see that they know their work, signs their time cards and tells them when to take time off. Of course we outline the work they do, check their results and write reports from their data. Normally we come into the lab as little as possible."

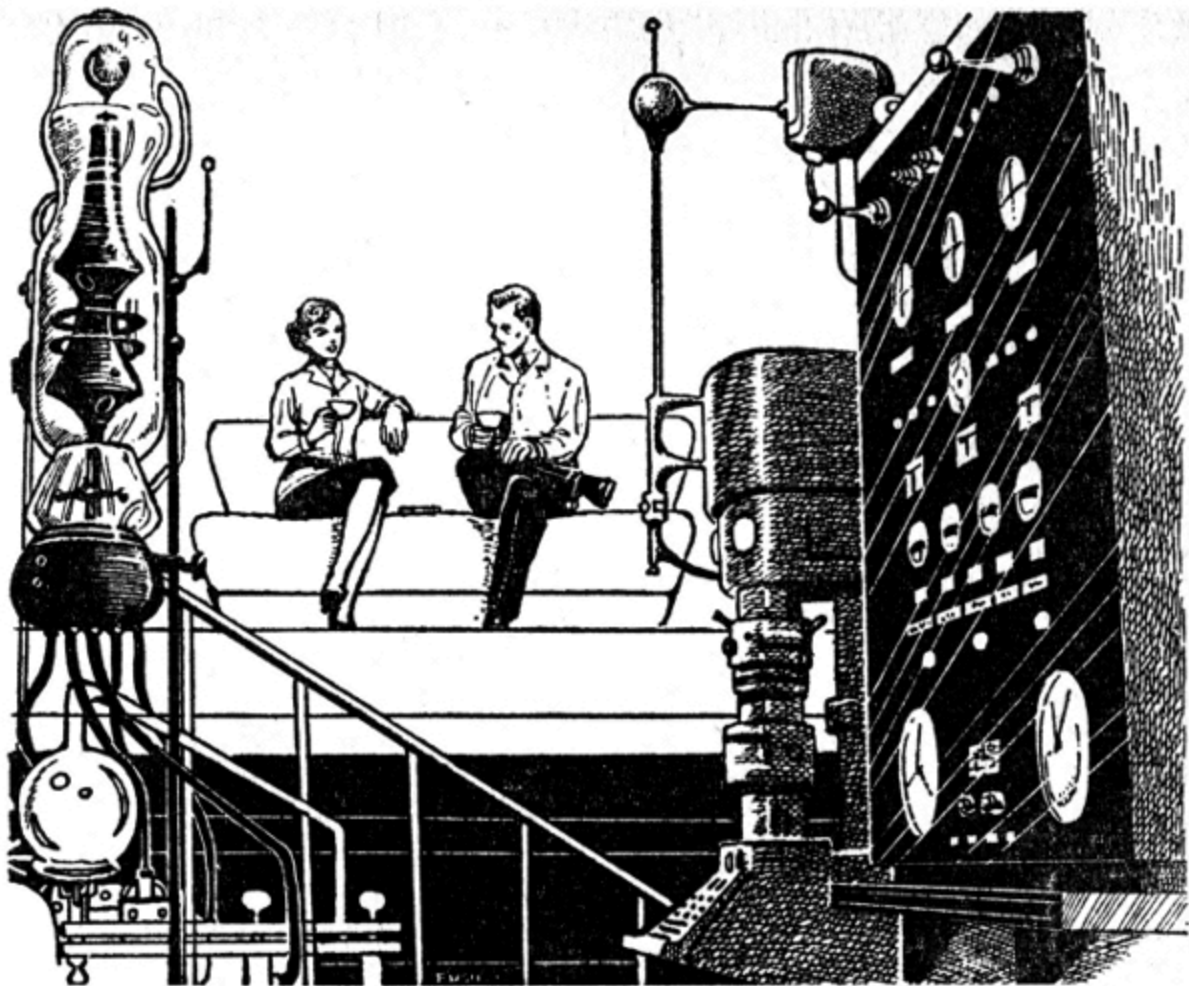
"But Sally, how the hell do we know that their results are right? This mixed-up outfit is in the hands of a bunch of left-handed prima donnas who don't know Beilstein from Budweiser!"

She smiled again (and he thought of the ads for Stargleam toothpaste). "Let's go over to the Scientists' Snack Bar and get a cup of coffee, and I'll tell you a little about the history of this laboratory."

So he let her lead him out of the individual laboratory into the pastel blue corridor where they followed the spiral runways to the glass enclosed Snack

Bar.

Here they sat on pale leather chairs and looked out over the expanse of the Central Laboratory. From where he sat, he could see a square mile of magnificent equipment: Serpentine condensers, enormous distillation columns, molecular stills, ultra-centrifuges, electron microscopes, all were spread out before him. Surrounding the central laboratory were the innumerable railings of the corridors leading to the individual offices. Upstairs and downstairs strolled scientists and Laboratorians respectively, all obviously contented. He turned to face Miss Chester who was lolling in the chair beside him. She had poured him a cup of coffee, given him a plate of rolls and was ready to talk.



She reminded him that in 2005 it was found necessary to build research laboratories on the Moon to avoid the guided meteorites which the Aliens had been hurling toward the Earth. Since there had also been a shortage of trained scientists, it was necessary to train apprentices to operate the complicated laboratory equipment ... to perform the operations without bothering themselves with the theory. The Laboratorians were needed and they did a good job running specification tests on all the equipment necessary for the interplanetary war. After the war, the Interspatial Corporation had made it the Central Research Laboratory, since this had been the largest aggregation of instruments ever gathered together, and in the ten intervening years, the numbers of college-trained scientists had increased almost ten-fold. As long as the Laboratorians confined their work to the equipment they were familiar with, they were unbeatable. To guide them they had the Book, as the Technical Manual of the Interspatial Corporation was known, and the Laboratorians followed its procedures to the letter.

"But they don't know *why* they're doing things," Whitemarsh interrupted. "The manual's been in need of revision for the last five years, and research workers don't use the same tests all the time!"

"Well that's right," admitted Sally without disagreement. "I usually have my particular laboratory instructions mimeotyped and bound in a little book. I've also got the instructions so fixed that if they do things wrong, I can catch them. And I've learned not to modify my instructions orally. That only confuses the men and results in chaos. With a little planning, you can get good work done, and if you don't mind humoring their whims a little, there's no reason why you can't get along with them."

Whitemarsh wasn't so sure. He had no objections to jollyng his subordinates, but he did draw the line at sloppy lab technique. He escorted Miss Chester to her own office, thanked her for the briefing, and then started to worry on his own. He took the speed elevator up to Dr. Sheridan's office.

The Laboratory director was sympathetic. He looked at the broad-shouldered young giant, Dr. Whitemarsh, and reflected that this man was

rated the most promising scientist the Interspatial Corporation ever had.

"You're damn right, Whitemarsh," he told the younger man, pushing him into a chair and offering him a cigarette. "I've been here three years and spent the first two fighting the system. Maybe the trouble goes back to our Board of Directors. They're all so proud of this shining Research Station on the Moon, that they hate to admit that anything's wrong. They've got the Laboratorians responsible to the Lunar Mines Service—and there it stands.

"So the only thing we can do is wait. Lo Presti the Master Mechanic is up for retirement next year and there's going to be a big organizational shake-up. Hold tight. After that we may have a free hand."

So Whitemarsh thanked him and bided his time. He released Rocco back to some other scientist and did his own laboratory work, even though the Laboratorian Council made a written protest. He also spent many hours in the excellent laboratory library, reading all the reports coming out of the Lunar Laboratory over the past ten years.

His discoveries amazed him. Theoretically the Lunar Lab had one of the best collections of scientific minds in the Solar System. Every Earth university was represented on its staff. New techniques and products had poured out of the Laboratory during the ten years of its existence, yet every one of these had been based on doubtful data. Certain things worried him. First, notes were kept in a very cavalier manner even by the most experienced scientists. Secondly, the younger chemists and physicists never had been exposed to any practical laboratory work after their student days, and consequently had no means of judging the technique of their assistants. Finally, the Laboratorians were apparently proud of their ignorance, displayed a contempt for "paper work" and were only too willing to fix their results if they thought they could get away with it....

He did not let his social development slide either. Lunarport was far more advanced culturally than the crude settlement on Phobus. Here Dr. Whitemarsh was able to have a luxurious apartment in the New Dome sector, could hear lectures and concerts, and could even indulge in winter sports such as skiing in the lava around the craters (protected of course by a heated suit and an oxygen mask.) He found Miss Chester a satisfactory companion for such endeavors, even though she spoke little of her private life or how she had avoided marriage in her twenty-five years. But he

played a waiting game with her as well as with the lab job. He admitted to himself that a research chemist's life at Lunar Lab was a pleasant one, particularly if one didn't care how accurate one's results were. Unfortunately, the same quirk which had driven him into science also made him suspicious of all easy methods. He had never recovered from the shock of discovering that just because a reaction worked in a book, it did not necessarily have to do so in a laboratory.

Dr. Whitemarsh's promotion came within five rather than six months. There was some grumbling among the older scientists, but there was not much they could do about it. Kercheval, who had twelve years' service on the Moon, did not have his Ph.D. and did not care particularly for executive work. Neither did Sturtevant with a doctorate and ten years service. But others objected; even Miss Chester, long one of Whitemarsh's defenders, felt that the older men deserved at least the chance of refusal. (It never occurred to Whitemarsh that she might have had some ambitions of her own.)

He called the group leaders together for a conference the day after his appointment. He was now ensconced behind Sheridan's desk and was not yet accustomed to having a secretary. The leaders came in grim and resentful. He wasted no words.

"I'm going to reorganize the set-up to get the Laboratorians under us, whether they like it or not. This sloppy technical data and unsubstantiated findings is not my idea of a good lab—nor yours, I'm sure. It's up to you to show it during the next year. Meanwhile you've all been pushed up fifty dollars a month in salary. So long!"

His next step was to call on Lo Presti. The Master Mechanic's Office was outside the Lab Dome near the Shaft of Lunar Mine No. 1. The old man had been in the preliminary Selenium exploration party and never could forget the old days when he drove the men and robots to find the metal that paid for the cost of the Expedition. The President of the Home Office, Dr. Barker, had never forgotten either, and Lo Presti was always taken care of. The 200 Laboratorians probably caused him more headaches than the five

thousand miners ever had, since a delegation visited him every day or so now that Dr. Whitemarsh was rumored in.

But the Lo Presti knew that times change too, and realized that the brawling space adventurer did not fit into a sleek world of test tubes and retorts. Ninety-five years old and arrogant as ever, he sat in his office and greeted Dr. Whitemarsh with a bonecrushing handshake. He offered a cigar and Whitemarsh thanked him, lighting a pipe instead.

"I hear from the boys you've been cracking down on them," he stated.

"No more than you would if you'd been there yourself. What would you do if a driller split a core?"

"Why I'd give the careless sap a clout that would wake him up. But the Laboratorians aren't drillers!"

"That's right, but that's the way some of them are muffing their work."

Lo Presti eyed him appraisingly. "Aren't you the same Whitemarsh who capped the crater on Phobus last year?"

"I sure am. And your Laboratorians are a bevy of Nice Nellies compared to that mutinous bunch of space rats I had with me."

"Well, maybe you're the man for the job at that. The guys don't put out anymore. Used to be I knew all the gang. I'd look around and see when they were goofing off. Now they're all such experts, I can't tell if they're loafing or just thinking." They both laughed at that. Whitemarsh thought it would be a good time to say: "I don't want to do anything to your boys for a while until I get my own gang straightened out!"

"Don't kid me, Doc," responded Lo Presti, "you know when I retire you're going to move in and crack down. Well I'm with you!"

So they parted friends.

Whitemarsh went back to his office in a happy mood. True, Miss Chester had been avoiding him lately and he had to drink coffee by himself but he now had the foremen on his side and the front office. Now was the chance to reform the laboratory.

His first bombshell was the requirement that all the junior chemists should take a qualifying examination. That really caused trouble in paradise.

Apparently, all of the younger set had thrown away their books on graduation and remembered only their own specialties. Whitemarsh, from being a pleasant companion at the Snack Bar who discussed skiing and spaceball, had now become an ogre of the first water. The senior chemists chuckled, since they were exempt, and the Laboratorians guffawed aloud to see their harriers in turn harried. In any event there was frenzied activity in the month before the examination and the library staff did yeoman duty. And, no one had threatened to quit. At least almost no one. Whitemarsh was musingly staring out of his office's Plastoid window at the green eye of Earth when he heard a commotion outside in the ante-room. He looked out to see Sally Chester, and he sensed that their relationship was less than idyllic.

"Let me see that egotistical ass, Whitemarsh," she shouted at his secretary who cowered in silk clad finery as the white-coated Valkyrie charged by.

"Be calm," he advised her, placing himself strategically behind his desk.

"Calm," she screamed, "how can I be calm when an officious busybody starts getting drunk with power and acting like a Twentieth Century dictator? After all I've done for this stinking Lunar Lab, how come that I have to take an exam in freshman chemistry?"

"I thought you were exempt," began the chastened director.

"Sorry, your honor! Your order says five years at Lunarport. I've only been around this sweat shop for four years and six months. What are you going to do if I fail? Throw me out and I'm moving over to Campo Sano with every one of our trade secrets!"

"I'll get you exempted," he offered.

"What, and have the other chemists cry favoritism? Not on your life, you coffee-swilling Judas," she yelled. "And stop grinning at me like a Cheshire Cat!"

He did not answer. He was content only to admire her in her rage. Her usually mild face was flushed through the tan and her graceful hands were tightly clenched into fists that pounded on his desk.

"Answer me, you moron!" she shouted. Then she started to cry. Within one minute the seething Amazon had changed into a defenseless white-coated

girl cowering in the visitor's chair, weeping bitterly. Whitemarsh approached and held her hand.

"Listen, Sally," he told her, "the only reason I was going to let you out of the test was because you know more chemistry than any of the scientists here. But go ahead and take the test; you'll get the highest grade!"

She brightened, "You think so?"

"Know it," he affirmed gallantly, "now, how about going to the Space Opera at the Symphorium tomorrow? Kluchesky is singing in *Pomme de Terre*."

She stiffened slightly and stood up. "Listen, Mr. Frank Whitemarsh! Privately you're not a bad guy. You even had potentialities. But you're a hell of a failure as a boss and the less I see of you, the happier I'll be. Good-bye!" And she was gone. Whitemarsh resumed his contemplation of the Earth with less interest.

The results of the examination might have been foretold. The intelligent and professionally alert junior chemists retained enough fundamentals to do well. The majority failed the questions on laboratory technique. Consequently Whitemarsh enlisted the aid of the older men to conduct a series of refresher lectures to bring up to date the scientific knowledge of those who failed. The Laboratorians were delighted with the spectacle presented by these lectures, and loved going home at night while erstwhile bosses sat listening to Dr. Sturtevant discuss "The Theory of Washing Precipitates", or to hear Dr. Whitemarsh talk on "Balancing the Redox equation." The Laboratorians' happiness lasted until one day in October.

That was the day that Lo Presti retired. The old man was given a small space ship by the Corporation and a space-time chronometer by the Laboratorians. Then he sorrowfully said farewell. The next day the Laboratorians were absorbed into Research.

Somebody had to plan for janitor service, figure where to place time cards, design new proficiency ratings and decide on such complex matters as where the Laboratorians were to hang their coats. All these services had been provided for by the miner's shop organization. Whitemarsh stayed late

at night for a week arranging the new payroll plan and raising the salaries somewhat.

All this was handled, if not without incidents, at least without violence. Even the janitors and secretaries were now part of a team. All but Miss Chester. She had stopped speaking to Whitemarsh in the halls and had been seen in the company of a younger (and Whitemarsh felt) better looking physicist.

Then Whitemarsh dropped his second bombshell. The junior chemists were ordered to rate the Laboratorians for proficiency! Fresh from six months' study under such taskmasters as Whitemarsh and Kercheval, the chastened scientists were now able to interpret the antics of their tormentors of yesterday. An old tradition had fallen and the howls extended back to the Front Office on Earth.

For a change, Miss Chester did not object. She was evidently past all comment. She merely wrote out a list of the faults and virtues of all her assistants, rated them all Excellent and went back to her research.

But Rocco was tried and found incapable of running titrations. Harry Crowe was found to be weighing incorrectly, Zachary had been fixing his calculations for the last ten years and even faithful Bruno had been found to be adding 15 to all of his Iodine numbers in order to pass the specs easier.

It suddenly occurred to every one that all the laboratory's reports were based on incorrect data. All work stopped for a week until the scientists found what their assistants had been trying to do all along. And the results were a bit terrifying. When Kercheval found that an incorrectly calibrated reflectometer had negated five years of his pet project, he tore up his notebooks, flung them on the floor and stalked into Whitemarsh's office.

"Frank, I'm taking my back vacations and going to Venus to forget it all for about six months. And mind you, when I get back I don't want to see my present assistants. I'm going to start from scratch."

He left, banging the door.

Next was Sturtevant.

"Frank, we've got to get Interstellar Review to hold my last paper. I want to recheck the melting points of some of those diazo compounds."

Then came the young physicist, Dr. Slezak, who was rumored to be Miss Chester's present skiing companion. "Dr. Whitemarsh," he stammered, "I'm not sure about the data on my last report."

"Didn't you take it all yourself?"

"Yes, but I used some of Kercheval's data for my fundamental calculations and, if that's wrong, all my conclusions may not be valid."

"Stop worrying," Whitemarsh told him. "When Kercheval recalculates his values, you can revise your own report. As long as your own work is right, you have nothing to worry about."

The young man left, nervously wringing his hands. Whitemarsh couldn't see what Sally saw in him. He figured she ought to be along by now.

She was.

"I told you so," Sally said theatrically. "You've got the whole lab mistrusting each other. All the chemists are quarreling like mad and the Laboratorians all look like whipped dogs. You've pulled the chair right out from under everything and you sit here gloating."

"Relax, Sally," he told her. "They're just growing pains. Take it easy and ride out the storm.... Now, how about tearing over to Lunar 7 to see the crucial Spaceball series between the Space Rangers and the Callisto Satellites?"

She looked horrified. "I'm afraid you don't take hints very well. I'm not interested in going anywhere with you. Actually, I'm going with Jack Slezak to see 'Nova of the Leprous Soul', and I might suggest a fit subject."

She flounced out again and Whitemarsh felt lost. He tried to cheer himself with a book on *Hyper Plutonium Elements*.

The transition took longer than Whitemarsh had bargained for. After the Laboratorians were re-educated, and a tiresome process it was, chemists went over the notebooks to look for inaccuracies, doubtful data was examined, all microfilms had to be edited and corrected; and they found that most of the chemicals developed at the laboratory in the past decade had been founded on doubtful data. But since all of them had passed the Development Group, Whitemarsh didn't think it was wise to try to recall

them. But new products scheduled for release were re-examined and retested after the fundamental work on them was checked.

Finally the problems were unscrambled and the laboratory began to run smoothly again. The research projects were reestablished and the work started out anew. Frayed tempers were soothed and the scientists finally got around to trusting each others' results again. The Laboratorians were now carefully but tactfully watched by the junior chemists who, in turn, were spending more time in the laboratories and less in their offices.

When the new, sound results started grinding forth, Whitemarsh permitted himself a sigh of relief. Lunar Lab had lost its individuality, he admitted, even though the easy-going camaraderie he had noticed when he first came was also gone. The results of Lunar Research Lab of Interspatial were now as reliable as those of the *Campo Sano* and *Roque* laboratories back on Earth.

But it had been a hard fight. None of the chemists ever stopped around his office any more for small talk about sports and politics. His secretary brought him coffee in his sanctum sanctorum and he did not find himself wandering around the laboratory as he had formerly done. When he did, there was usually a restrained silence and a suspicious neatness. Miss Chester was apparently irrevocably lost and there were rumors of an engagement with the brilliant Dr. Slezak. Though he had won the day, he had lost something too. The Lab was now able to turn out results, but Frank Whitemarsh had paid a personal price for its new efficiency.

Almost a year after taking over as Research Director, Sheridan, now a Vice President, brought him some news. "Get ready to pack, Frank," he told the younger man as they sat and smoked in the director's office watching the clouds moving over the Earth.

"The Front Office like what I did?" asked Whitemarsh puffing on his pipe.

"Well." There was a slight pause. "All the scientists on the board are behind you to a man. But the business men, the advertising boys and accountants, well ... you know how they are."

"What's eating them?"

"The lab didn't release any new products this past year. Development and even Advertising are pretty much slowed down."

"That's right. We've got some good products about ready, but we're making a final check before release. Don't you think we sent out a lot of junk before?"

"We sure did, even in my time though I tried to stop it. But the development boys want something, anything."

"Well?" asked Whitemarsh.

"So they'd probably rather run the risk of getting something bad than nothing at all."

"They won't!"

"That's right, they never will again. Now, I know that the products you have ready are going to be good and I'm not worried about them. All we have to do is keep the business geniuses out of our hair for another six months."

"And?"

"So we're kicking you upstairs. It's a good job, don't worry about that, at three times your director's salary."

"What if I quit?"

"Don't be that silly."

"What's the other job?"

"Works Manager at Quercus Mountain on Phobus. Sole boss of the biggest Isotope Works in the Solar System. You'll have 50,000 men under you and have a free hand at starting any kind of laboratory you want."

"No Laboratorians?"

"Right. You can start out from scratch and make the kind of lab you've always dreamed of. Here we're thinking of pushing up Kercheval if it's all right with you, you always rated him highly. It's just like changing Spaceball managers. We all know the Space Sox won the pennant last year on the team developed by Kanter even though Balhiser was manager. These

wolves will keep off our tail until the new products start coming through and then we'll say we knew it all along."

"You've got me half convinced not to quit," said Whitemarsh quietly.

"Now listen Frank," came back Sheridan just as seriously, "you're too good a man to waste. Now take your promotion like a nice boy and keep in line."

"I still think I did a good job here."

"So do I, but the Board of Directors can't forgive those retractions, even though you and I know they're necessary. They don't know what scientific truth and pride are. Within ten years, on the foundations you laid, we'll have the best research record in the country...."

After Sheridan had left, Whitemarsh cast a last look at his former domain. He called Kercheval in to give him the news and then tell him to keep quiet until verified. Then he decided to take a last tour around the laboratories. He finally found himself up at the Snack Bar and his eyes were taking the same look over the Laboratory that they had done two years before. The view looked about the same. He had supervised the installation of a new Matter Probe over in the front center and he was responsible for the Atom Analyzer, but these were only minor changes.

The major change, he thought bitterly, is that no one speaks to me unless spoken to—I've become a pariah. Never tamper with the status quo, it disturbs too many people. It's a very lonely job.

There was no one else in the Snack Bar. At least, almost no one else. He heard a discreet cough behind him. He turned and found Miss Chester seated behind him. She had her legs crossed, a cup of coffee in one hand and the Space News Want-Ads in the other.

"Hello, Napoleon," she greeted him. "Have you just been surveying your empire? Did you see the stern men of science jumping through the hoops out there? Can you remember the happy place this was a year ago when you came? Then the Laboratorians took pride in their work; now they're flunkies for the green kids fresh from Alma Mater!"

"Stop it, Sally," he told her. "You're not too far wrong on that Napoleon business. I'm taking off for my new St. Helena, Quercus Mountain on Phobus."

"Quercus Mountain? That's a big place. Lab Director?"

"No. Works Manager."

"Heaven help the poor Atomic workers!"

"Don't be that harsh. Dammit! Sally, maybe I am a Napoleon, but scientific accuracy is too important to play fast and loose with, the way they were around here. You know it. You're the only one who didn't relax that vigilance—who saw to it that everything you turned out was without error. I know now that I forgot the human equation—that I was so eager for errorless research that I trod pretty roughshod over a lot of people. But you're guilty too, you know, you had the secret—you managed to balance the equation when everyone else here didn't. Why didn't you help me? Sure, you came in and ranted and raved at me—called me all sorts of names, but you didn't help me, you didn't try to show me the way."

"I—"

"Let me finish," he interrupted her. "I love you, you know—have for a long, long, time. I still need help, Sally. I don't want to keep playing Napoleon and going into exile over and over again. A bigger job with more men under me isn't the answer. When a man is lonely it makes him hard and cruel in circumstances like that. I made all of you here relearn scientific facts, I need to relearn the humanities...." He paused for a moment. "Sally, will you teach me?"

Her eyes were bright with unshed tears and a catch in her throat made the words husky and half-whispered. "I wanted to help—I love you too—but I thought you were arrogant and didn't need me—" She swallowed, controlling a sob. "I'll make it up to you, darling. You won't be alone again—on Phobus or anywhere else in the galaxy."

*** END OF THE PROJECT GUTENBERG EBOOK THE
LABORATORIANS ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.